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RESEARCH ARTICLE



Supply and demand prediction analysis of urban gas energy in China: A new method using WOA–Grey Markov

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ABSTRACT

Urban gas supply–demand forecasting provides a critical basis for formulating policies on energy security, infrastructure planning, and low-carbon transition. This study develops a Grey–Markov model enhanced by the Whale Optimization Algorithm (WOA–GM–Markov). Using Chinese urban statistical data from 2014 to 2023, the model fits and forecasts the supply volumes and user populations of natural gas, artificial gas, and liquefied petroleum gas. The proposed approach is compared with GM(1,1), GM–Markov, ARIMA, and PSO-optimized GM–Markov. Results show that WOA–GM–Markov achieves optimal accuracy across all sequences. For natural gas users, RMSE is reduced by approximately 97.1% relative to ARIMA, and for artificial gas supply, RMSE is reduced to 3.26. Rolling-origin validation with a fixed six-year training window yields out-of-sample RMSE ratios ranging from 59.9 to 194.1% relative to full-sample fitting, confirming the model's stability under strict temporal constraints. Comparisons with PSO-optimized variants demonstrate that WOA achieves marginally superior precision on select sequences while eliminating the performance degradation risk inherent in subjective parameter selection. The proposed model effectively overcomes the subjectivity of traditional GM–Markov and provides quantitative decision support for urban energy management and differentiated policy design under dual-carbon goals.

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Energy supply–demand prediction; GM–Markov model; low-carbon energy policy; urban energy forecasting; Whale Optimization Algorithm (WOA)

1. Introduction

To meet the growing energy demand of urban populations, urban energy planning plays a crucial role (Vakilifard et al. 2019). As the core carriers of regional economic and social development, cities need to reach a balance between their high-quality development and stable operation of energy systems. Primary energy constitutes a significant component of current urban energy consumption, and the development of its supply and demand directly determine the normal operation of urban industrial activities, residents' livelihoods, and public services.

Urban development drives the evolution of primary energy supply–demand relationships through population agglomeration and industrial upgrading. From the demand side, the continuous inflow of population amid urbanization directly expands the scale of residential energy consumption, e.g., heating and transportation. Changes in national and regional population structures (Huang, Lin, and Guo 2024) and regional economic levels (Wang et al. 2024) have further demands for high-quality energy services. Meanwhile, despite the gradual shift toward low-carbon development in industrial transformation and upgrading, including technological upgrading of heavy industries (Li 2025), digital transformation (Jing et al. 2025), and the rise of emerging manufacturing sectors (Mallalieu et al. 2024), the necessity of short-term prediction on primary energy remains strong (Pan, Liu, and Li 2018; Stoner et al. 2021; Yunusa et al. 2023).

The formulation of national energy support policies for urban development requires macro-level data forecasting and research. Effective policymaking, in turn, must be based on scientific predictions. Forecasting the supply and demand of primary energy is therefore a critical research endeavor for the implementation of urban energy policies. The contribution of our study are as follows:

- (1) Energy supply planning. By forecasting energy demand of national urban areas (short-term), we provide academic support to policymakers, thus adjusting energy policy correspondingly.
- (2) Development demand management. Combining relevant data from official website to forecast the energy-consuming population in urban areas, laying foundation for future population absorption and urban expansion, ensuring energy safety and System resilience (Cohen 2025).
- (3) Preparation for low-carbon transition. As many countries will approach their carbon peaking in the future (Zhu et al. 2025), short-term forecasting of primary energy supply and demand can provide research support for the development of sustainable energy.

All of our work will help strike a balance between ensuring energy supply for development needs and reducing carbon emissions from primary energy.

Forecasting primary energy supply and demand has become a major research direction. While China will further enhance its low-carbon and environmental protection efforts in the future, primary energy consumption and demand remain substantial. Predicting and understanding primary energy consumption and demand in the coming years, especially data related to urban areas of a country (Yang and Qin 2026) is helpful in improving energy use efficiency.

The structure of the article is as follows. Section 2 presents the research literature on the relevant methods. Section 3 provides a detailed description of the specific methods. Section 4 conducts research and discussion based on the data. Section 5 further verifies the model and do the sensitivity analysis. Section 6 is the conclusion.

2. Related work

2.1. Prediction cases in energy

Energy development and policy formulation require data support. Forecasting energy demand has emerged as a research direction currently. For large datasets, machine learning (ML) is typically employed to explore the intrinsic patterns of fluctuating data, with training and test sets used to validate the accuracy of methods. Moreover, comparisons among multiple ML approaches are commonly conducted. Wang et al. proposed a real-time electric load forecasting model within unsupervised learning, and demonstrated that a backtracking optimizer could enhance forecasting efficiency (Wang, Yao, and Papaefthymiou 2023). Choi et al. established a supply–demand management system integrating production, transportation, and utilization, and improved forecasting accuracy using reinforcement learning (RL), yielding economic benefits (Choi et al. 2025). B. Vijay et al. developed an energy management and optimization system, where the SEMS-IOT-MFRCNN exhibited excellent predictive performance (Vijay Kumar 2025). Su et al. employed a hybrid K-means and KNN machine learning approach to forecast short-term energy loads, incorporating holiday effects to address special scenarios such as major festivals (Hnin et al. 2024). A more common ML method is LSTM, which is widely applied in time series forecasting and has been utilized in energy subfields including national energy systems (Chaturvedi et al. 2022; Miah et al. 2024), buildings (Nygård et al. 2025), smart grids (Khayat et al. 2025; Shawon et al. 2025), and energy storage (V and K 2025).

Notably, all ML methods invariably require large volumes of data. Researchers often combine ML to improve prediction accuracy and validate stability of models through empirical studies. Application of different prediction methods are clarified (Mustafa and Al-Yozbaky 2025). For discrete time series data, the grey prediction model (namely GM) is simple to use, so numerous studies utilizing GM have considered data periodicity. Zhou et al. incorporated seasonal factors and employed time series with different cycle lengths and unit times to forecast China's import demand for petroleum, natural gas, and crude oil (Zhou et al. 2025). Xie et al. adopted a discrete variable grey prediction method to forecast China's total energy consumption, and used a novel Markov approach to predict trends in energy demand structure (Xie, Yuan, and Yang 2015). There are also many other studies on energy supply–demand forecasting using seasonal GM (Jiang et al. 2025; Saponken et al. 2026), as well as extensive research on innovations in grey theory (Ma et al. 2025; Shi and Gu 2025; Wang et al. 2025).

ARIMA and its variants are also used for forecasting seasonal or discrete small-sample data. An early study in 2007 applied ARIMA to forecast primary energy demand in Turkey, revolutionizing previous simplistic econometric model-based predictions (Ediger and Akar 2007). For seasonally varying data series, Spyridon et al. proposed a hybrid forecasting method combining Fourier decomposition and ARIMA to study the Greek energy market, verifying its superiority over traditional SARIMA in terms of MAPE, RMSE, and MAE (Karamolegkos and Koulouriotis 2025). More studies have focused on integrating ARIMA with other methods (A, Christo, and Elizabeth 2025) or ML approaches (Zhang et al. 2025).

Numerous contemporary studies have confirmed that hybrid methods combining traditional approaches and ML can leverage the strengths of each side, resulting in improved accuracy compared with traditional methods.

2.2. The combination of methods

Previous sections have summarized existing energy forecasting methods. However, short-term supply-demand forecasting methods for primary energy in urban areas still have certain limitations: Grey Models (GM) are suitable for poorly informative data, but their overly simplistic fitting leads to significant errors. Markov models excel at handling random fluctuations but rely on state division, which is highly subjective.

This study combines GM with Markov forecasting, a commonly used hybrid forecasting approach (Li, Yamaguchi, and Nagai 2007; Yu et al. 2022) to address the prediction errors. Furthermore, parameter setting in calculations has often lacked in-depth exploration, with parameters typically assigned directly. Herein, we introduce the Whale Optimization Algorithm (WOA) to search for and optimize relevant parameters, aiming to identify the optimal parameter that maximizes prediction accuracy. Meanwhile, the primary energy in this study covers multiple subcategories (natural gas, artificial gas, and liquefied petroleum gas), and comprehensive data forecasting is conducted to encompass major primary energy types common in urban areas.

3. Methodology

In this section, we elaborate on the mathematical methods employed: Grey Prediction Model (GM), Markov Chain, and Whale Optimization Algorithm (WOA). As outlined earlier, these methods are integrated for forecasting purposes. Grey-Markov (or GM-Markov), a common forecasting method, essentially involves predicting both original sequence and the residual sequence (Fu et al. 2021; Jia et al. 2020).

Traditional Grey-Markov operates as follows. GM is firstly used to generate initial predictions. Then, a set of residual sequences is derived. These residuals are predicted within Markov chain to determine residual state (a range which the residual belongs to), yielding a new residual value. This residual is ultimately used to correct the initial predictions from GM. Parameter adjustment is performed in the residual correction step (see Figure 1 below) with WOA (an optimization method, which will be introduced below).

3.1. Grey prediction model

The grey prediction model is a traditional forecasting method. Its core lies in representing the trend of a time series using an exponential function, making it suitable for small-sample, information-poor systems. In this study, applying this method to predict the next 3 years based on 10 years of historical data trends is highly reasonable.

Step 1 The original sequence is $X(0) = \{x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(k), \dots, x^{(0)}(n)\}$ where $k = 1, 2, \dots, n$, and the superscript (0) indicates the original sequence. Prior to conducting the grey prediction, a ratio test is required. This involves calculating the ratio of consecutive data points in the sequence as $c_k = \frac{x^{(0)}(k-1)}{x^{(0)}(k)}$, $k = 2, 3, \dots, n$. To verify the suitability of the sequence for grey prediction, the following conditions must be satisfied.

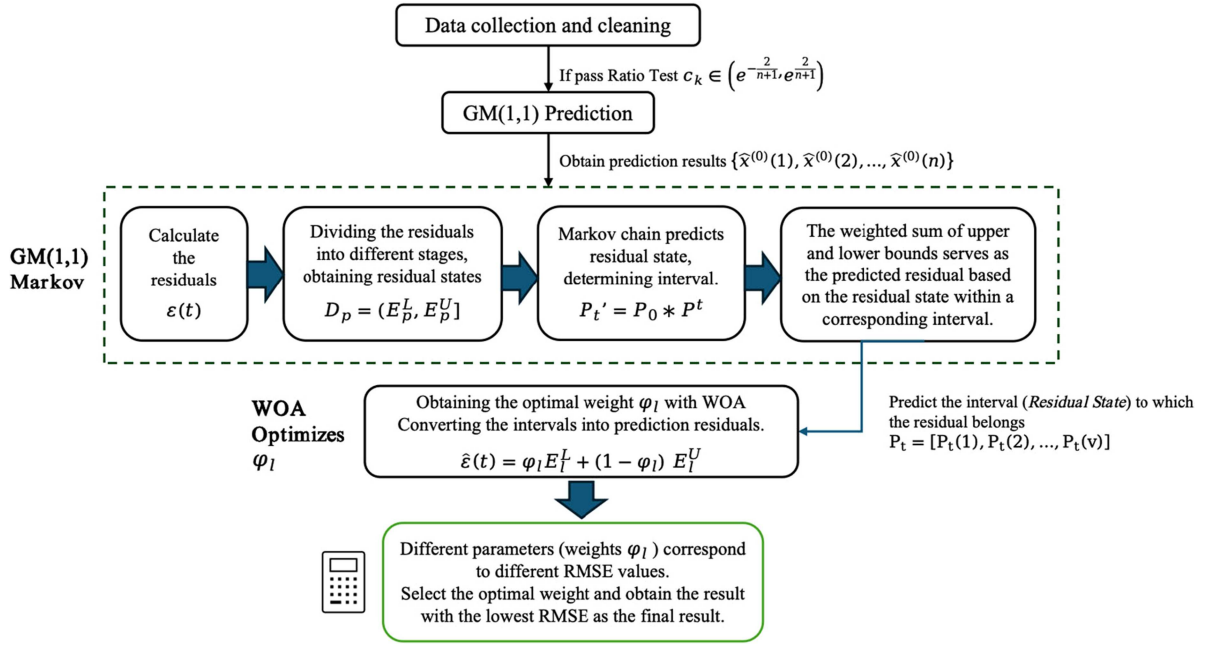


Figure 1. Flow chart of GM–Markov–WOA method.

$$c_k \in \left(e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}} \right). \quad (1)$$

Since energy supply and demand for 2024–2026 are forecasted based on data from 2014–2023, so the sample size $n = 10$. The following several steps provide a detailed explanation of the grey model.

Step 2 Generate the First-Order Accumulated Generation Operation (1-AGO) sequence, denoted as $X(1) = \{x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(k), \dots, x^{(1)}(n)\}$ – X is defined as the cumulative sum of the first k terms of the original sequence $X(0)$.

$$x^{(1)}(k) = \sum_{i=1}^k x^{(0)}(i). \quad (2)$$

Meanwhile, based on the $X(1)$ sequence, an adjacent mean sequence Z is generated, where $Z(1) = \{z^{(1)}(2), z^{(1)}(3), \dots, z^{(1)}(k), \dots, z^{(1)}(n)\}$ – Z is calculated as the average of the $(k-1)$ -th and k -th terms of X , applicable for $k = 1, 2, \dots, n$.

$$z^{(1)}(k) = \frac{1}{2}(x^{(1)}(k-1) + x^{(1)}(k)), \quad k = 1, 2, \dots, n. \quad (3)$$

Step 3 Construct the whitening differential equation of the GM(1,1) model using the sequences defined above. This differential equation takes the form of a first-order linear equation, where a represents the development coefficient and b denotes the grey action quantity. The unbiased estimators of a and b (denoted as $\hat{a}$ and $\hat{b}$ respectively) can be obtained using the least squares method (LSM).

$$\frac{dx^{(1)}}{dt} + ax^{(1)} = b. \quad (4)$$

The calculation follows a standard matrix form: the transpose of the parameter vector $[\hat{a}, \hat{b}]^T$ is linked to matrix B , Y .

$$[\hat{a}, \hat{b}]^T = (B^T B)^{-1} B^T Y. \quad (5)$$

Here, matrices (1) B and Y are constructed based on the adjacent mean sequence Z and the original sequence $X(0)$ respectively, following the structure specified for GM(1,1) modeling.

$$B = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \vdots & \vdots \\ -z^{(1)}(N) & 1 \end{bmatrix}, Y = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(n) \end{bmatrix}. \quad (6)$$

Step 4 The time response function of the GM(1,1) model is derived from solving the whitening differential equation.

$$\hat{x}^{(1)}(t) = \left[x^{(1)}(0) - \frac{\hat{b}}{\hat{a}} \right] e^{-\hat{a}(t-1)} + \frac{\hat{b}}{\hat{a}}. \quad (7)$$

This function describes the evolution of the 1-AGO sequence $X(1)$ over time. Notably, the above-derived function corresponds to the traditional GM(1,1) model; to obtain parameters for the unbiased GM(1,1) model, we can use the parameter values from the traditional GM(1,1) approach. Assume the original data sequence follows an exponential distribution.

$$x^{(0)}(k) = \lambda e^{\mu(k-1)}, \quad k = 1, 2, \dots, n. \quad (8)$$

where λ and μ are the parameters to be estimated. The 1-AGO sequence of this exponential sequence is calculated using the cumulative sum of the exponential terms. Furthermore, according to the established formula (Eq. (6)), the relationship between $\hat{a}$, $\hat{b}$ (from traditional GM(1,1)) and the estimated values of μ , λ (denoted as $\hat{\mu}$, $\hat{\lambda}$) can be derived. Specifically, $\hat{\mu} = \ln \frac{2-\hat{a}}{2+\hat{a}}$, $\hat{\lambda} = \frac{2\hat{b}}{2+\hat{a}}$.

Step 5 The fitted sequence of the original sequence, denoted as $\hat{X}^{(0)}$, is obtained through inverse transformation (reversing the 1-AGO operation) based on the time response function of the GM(1,1) model.

$$\begin{cases} \hat{x}^{(0)}(1) = x^{(0)}(1), & k = 1 \\ \hat{x}^{(0)}(k) = \frac{2\hat{b}}{2+\hat{a}} e^{\ln \frac{2-\hat{a}}{2+\hat{a}}(k-1)}, & k = 1, 2, \dots, n. \end{cases} \quad (9)$$

3.2. Markov correction and Grey-Markov

A Markov chain is a state-forecasting method characterized by the property that the state at time $t+1$ (denoted as s_{t+1}) depends solely on the state at time t (denoted as s_t) and is independent of all preceding states. This property can be mathematically expressed as:

$$P\{s_{t+1} = x_{t+1} | s_t = x_t, s_{t-1} = x_{t-1}, \dots, s_0 = x_0\} = P\{s_{t+1} = x_{t+1} | s_t = x_t\}. \quad (10)$$

The core rationale of the Grey-Markov method lies in addressing the significant prediction errors of the GPM and its poor fitting performance with the original data. To mitigate this issue, after obtaining the GPM-predicted sequence $\hat{X}(0) = \{\hat{x}^{(0)}(1), \hat{x}^{(0)}(2), \dots, \hat{x}^{(0)}(n)\}$, and calculate the residual ratio $\varepsilon(t)$ where $\varepsilon(k) = \frac{x^{(0)}(t) - \hat{x}^{(0)}(t)}{x^{(0)}(t)}$. From this calculation, the residual ratio sequence $\varepsilon = \{\varepsilon(1), \varepsilon(2), \dots, \varepsilon(t), \dots, \varepsilon(n)\}$ is generated.

Then define the relative error state space. The relative error sequence exhibits inherent regularity, which allows for correction using a Markov model (Li et al. 2022). Partition the residual ratio sequence ε into ν intervals (corresponding to ν states). For the p -th interval (where $p = 1, 2, \dots, \nu$), there exist a lower bound and an upper bound, denoted as $D_p = (E_p^L, E_p^U]$.

Construct the state transition matrix. First, classify each data point in the residual ratio sequence ε into one of the states (intervals D_i , $i = 1, 2, \dots, \nu$). Calculate N_i , the total number of residual ratios falling into

interval D_i and then count N_{ij} , the number of one-step transitions from interval D_i to interval D_j . The one-step transition probability from state i to state j is given by:

$$P_{ij} = \frac{N_i}{N_{ij}}. \quad (11)$$

With v intervals corresponding to v states, a $v * v$ state transition matrix P is constructed as follows:

$$P = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1v} \\ P_{21} & P_{22} & \cdots & P_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ P_{v1} & P_{v2} & \cdots & P_{vv} \end{bmatrix}_{v * v}. \quad (12)$$

For residual interval, the v states form an initial matrix $P_0 = [P_0(1), P_0(2), \dots, P_0(v)]_{1 * v}$, which satisfies two conditions: $\sum_{p=1}^v P_0(p) = 1$ and $P_0(p) = \{0, 1\}$. If $P_0(p) = 1$, it indicates that the residual ratio belongs to the p -th interval when $t = 0$. At the next time step ($t = 1$), updated probability matrix P_1 is calculated as:

$$P_1' = P_0 * P. \quad (13)$$

where $P_1' = [P_1(1)', P_1(2)', \dots, P_1(v)']_{1 * v}$. Select the maximum value in the sequence P_1' to determine the interval to which the next state belongs. Specifically, when $P_1(u)' = \max \{P_1(1)', P_1(2)', \dots, P_1(v)'\}$, set $P_1(u) = 1$ while all other elements being 0, thus obtaining $P_1 = [P_1(1), P_1(2), \dots, P_1(v)]_{1 * v}$. The matrix also satisfies $\sum_{p=1}^v P_1(p) = 1$ and $P_1(p) = \{0, 1\}$, $p = 1, 2, \dots, v$. For time t , probability matrix is:

$$P_t' = P_0 * P^t. \quad (14)$$

The corresponding state matrix $P_t = [P_t(1), P_t(2), \dots, P_t(v)]$ is derived from P_t' , where $P_t(l) = 1$ (indicating the residual ratio belongs to the l -th interval) and all other elements are 0.

Finally, correct the error. The interval corresponding to the l -th state is $D_l = (E_l^L, E_l^U)$. Forecast the residual ratio $\varepsilon(t)$, with the predicted residual $\hat{\varepsilon}(t)$ typically set to the midpoint of the interval D_l :

$$\hat{\varepsilon}(t) = \frac{E_l^L + E_l^U}{2}. \quad (15)$$

After obtaining the corrected residual, adjust GM-predicted sequence $\hat{X}^{(0)} = \{\hat{x}^{(0)}(1), \hat{x}^{(0)}(2), \dots, \hat{x}^{(0)}(t), \dots, \hat{x}^{(0)}(n)\}$ using the following formula to get the final corrected prediction value:

$$\hat{x}(t) = \frac{\hat{x}^{(0)}(t)}{1 - \hat{\varepsilon}(t)}. \quad (16)$$

3.3. Whale optimization algorithm

The Whale Optimization Algorithm (WOA) is a novel optimization algorithm inspired by the hunting behavior of humpback whales in the ocean (Madhusudanan and Padhmanabhaiyappan 2025; Pan et al. 2025). The coordinate position of each whale corresponds to a potential feasible solution to the problem. This mathematical model mainly simulates three types of behavioral patterns of whales: random search, encircling prey, and attacking prey (Figure 2).

Step 1 Simulate the random search behavior of whales in the ocean, where individuals explore the distribution by randomly selecting a position X_{rand} . The implementation is as follows:

$$X_{t+1} = X_{\text{rand}} - A \cdot D, \quad (17)$$

$$D = |C \cdot X_{\text{rand}} - X_t|. \quad (18)$$

X_t represents the new coordinate of the individual after position update in the t -th generation. D measures the distance between current individual X and randomly selected individual X_{rand} . C is a randomly

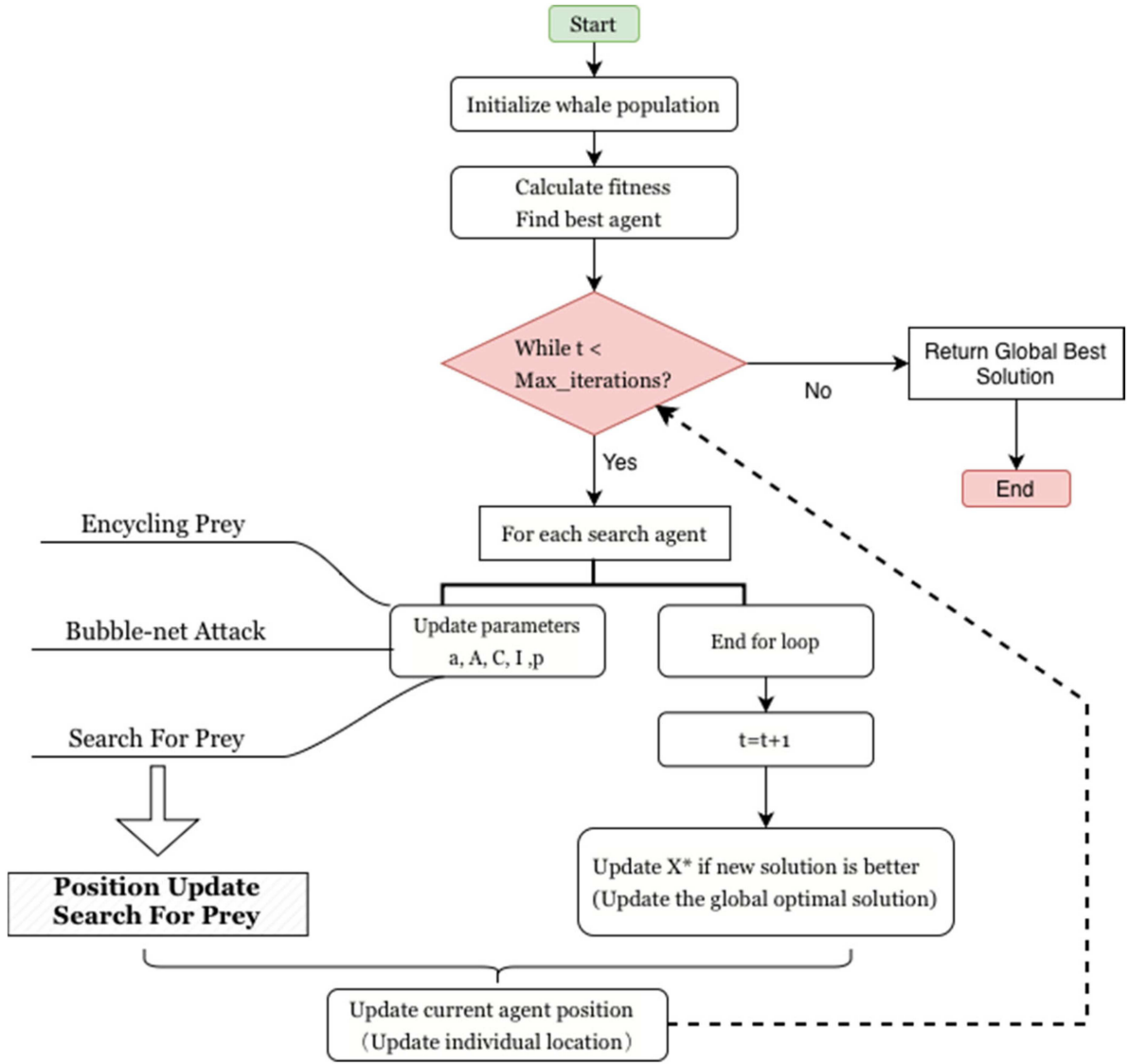


Figure 2. Flowchart of WOA.

generated value within $[0,2]$, which regulates the influence of the distance between X_{rand} and X on the individual's position update. The definitions of A and C are:

$$A = 2a \cdot r - a, \quad (19)$$

$$C = 2 \cdot r, \quad (20)$$

$$a = 2 - 2 \cdot \frac{t}{T_{\max}}. \quad (21)$$

where t denotes the current number of iterations. r is a random vector with values in $(0,1)$. a is a dynamically changing parameter that linearly decreases from 2 to 0 during the iteration. $T_{\max}$ represents the maximum number of iterations.

Step 2 Since humpback whales in the ocean not only approach prey in a spiral trajectory but also gradually shrink the encirclement around the prey during hunting, the algorithm updates position by selecting one of the two strategies with 50% probability.

$$\begin{cases} X_{t+1} = X_{\text{best}} - A \cdot |C \cdot X_{\text{best}} - X_t| & \text{if } p < 0.5 \\ X_{t+1} = X_{\text{best}} + D_p \cdot e^{bl} \cdot \cos(2\pi l) & \text{if } p > 0.5 \end{cases} \quad (22)$$

$D_p = |X_{\text{best}} - X_t|$ characterizes the spatial distance between individual X and current optimal individual X_{best} . b is a preset constant that determines the shape of the spiral trajectory. l is a randomly generated value within $[-1, 1]$. When $l = -1$, it means the artificial whale is closest to the prey (food). When $l = 1$, it indicates the artificial whale is farthest from the prey (food).

A is a random vector within $[-2, 2]$. When $|A| > 1$, it causes whale individual to deviate from the prey's position, thereby searching for other areas with more potential prey, which enhances the algorithm's global exploration capability. When $|A| \leq 1$, the whale individual moves toward the current optimal whale position.

3.4. WOA optimization parameters

According to Eq. (15), the residual is determined by the upper and lower bounds of the interval $D_l = [E_l^L, E_l^U]$ corresponding to the predicted state. Generally, the midpoint of the interval is adopted. However, we consider that such direct assignment is not an optimal or rigorous determination method.

Therefore, we introduce WOA to search for the value, aiming to derive a potentially better solution. We assume the weight is φ_l , so the predicted residual $\hat{\varepsilon}(t)$ satisfies.

$$\hat{\varepsilon}(t) = \varphi_l E_l^L + (1 - \varphi_l) E_l^U. \quad (23)$$

Our goal is to search for the parameter φ_l to obtain the optimal performance.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{k=1}^n (\hat{x}^{(0)}(k) - x^{(0)}(k))^2}, \quad (24)$$

$$\text{MAE} = \frac{1}{n} \sum_{k=1}^n |\hat{x}^{(0)}(k) - x^{(0)}(k)|. \quad (25)$$

The above two metrics are frequently used for the evaluation of prediction accuracy, with smaller values indicating higher accuracy.

4. Case study

Data on urban energy supply and demand were selected from the database of the National Bureau of Statistics of China. The original dataset includes the supply volume of primary energy in Chinese urban areas from 2000 to 2023. Since this study adopts a small-sample forecasting model, a recent 10-year period (2014–2023) was selected, covering three energy types: Natural Gas (NG), Artificial Gas (AG), and Liquefied Petroleum Gas (LPG) (C. Yang, Zhou, and Jackson 2014), along with corresponding Energy Supply and User demand data. (Table 1).

4.1. Pre-work for GM–Markov

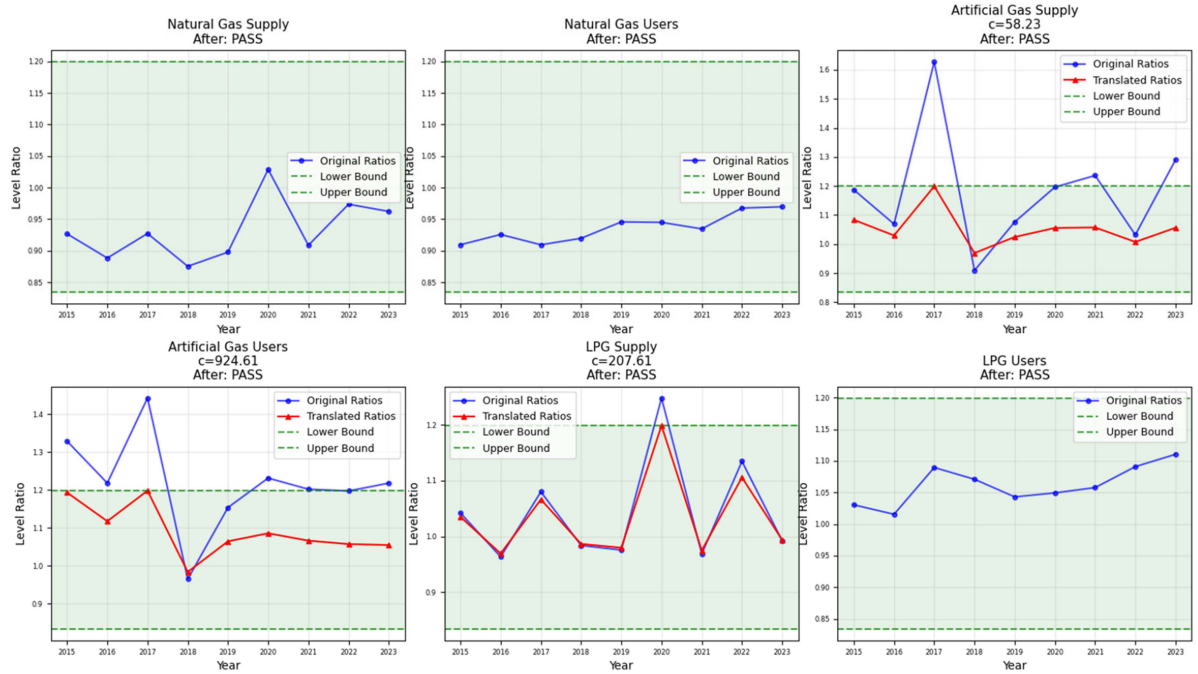
Referring to Eq. (1), ratio test is a pre-work for forecasting time series data, only data that pass the test can be used. Through calculations, we found that some of the 6 sequences failed to pass the test. The ratio values of each sequence are presented below. For sequences that failed the test, an optimal translation constant move needs to be calculated. Specifically, a range of move is derived by ensuring the ratio values meet the test criteria, and then the minimum value within this range is selected as move_{opt} .

$$c'_k = \frac{x^{(0)}(k-1) + \text{move}}{x^{(0)}(k) + \text{move}} \in \left(e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}} \right), \text{ for } k = 2, 3, \dots, n. \quad (26)$$

Table 1. Primary energy supply and users in urban area.

Year	Natural gas supply (10^8 m ³)	Natural gas users (10^4 people)	Artificial gas supply (10^8 m ³)	Artificial gas users (10^4 people)	LPG supply (10^4 tons)	LPG users (10^4 people)
2023	1837.19	47114.77	14.06	312.3	764.6	8405.8
2022	1767.7	45678.51	18.14	380.51	758.46	9332.83
2021	1721.06	44195.53	18.72	455.77	860.68	10180.49
2020	1563.7	41302	23.14	548	833.71	10767
2019	1608.56	39025	27.68	675	1040.81	11297
2018	1443.95	36902	29.79	779	1015.33	11782
2017	1263.75	33934	27.09	752	998.81	12616
2016	1171.72	30856	44.09	1085	1078.8	13744
2015	1040.79	28561	47.14	1322	1039.22	13955
2014	964.38	25972.94	55.95	1757.01	1082.85	14378.41

Data sources: <https://data.stats.gov.cn/easyquery.htm?cn=C01>.

**Figure 3.** Grade ratio test (c is $move_{opt}$).

After determining the range $(e^{-\frac{2}{n+1}}, e^{\frac{2}{n+1}})$, the minimum value of c'_k in this range is set. The ratio test is repeated until the adjusted sequences pass the test. The following is a comparison between the ratio values of the original data and the adjusted data: NG Supply, NG Users, and LPG Users passed the test, while the remaining sequences required adjustment (Figure 3).

After adjusting the data sequences, GM(1,1) was applied. The 1-AGO sequence was generated and the whitening coefficient and grey action quantity were obtained by solving the differential equation. Finally, inverse 1-AGO operation (i.e., reversing the accumulation) was performed to derive the forecasting sequence. The forecasting results of GM(1,1) are presented below (Table 2), including both the results of adjusted (after translation) and GM-predicted data.

Then, based on the above data, we calculated the residuals and residual ratios for the 6 groups (see Section 3.2), which are summarized in Table 3 below.

Following steps of GM, we next divided residual ratios into different interval. Here, equal-difference division (from the minimum to the maximum of residual ratios) was adopted, and division intervals are shown below in Table 4.

In the table above, D_p ($p = 1, 2, 3$) represents that we have three division intervals and their corresponding states. The upper and lower bounds of the intervals will be used in subsequent steps. Here, the residual

Table 2a. Prediction result of GM(1,1) for energy supply.

Years	Natural gas (10^8 m^3)		Artificial gas (10^8 m^3)		LPG (10^4 tons)	
	Actual value	Predicted value	Actual value	Predicted value	Actual value	Predicted value
2014	964.38	964.38	55.95	55.95	1082.85	1082.85
2015	1040.79	1132.99	47.14	44.43	1039.22	1103.56
2016	1171.72	1209.45	44.09	39.8	1078.8	1057.11
2017	1263.75	1291.06	27.09	35.38	998.81	1012.32
2018	1443.95	1378.19	29.79	31.16	1015.33	969.11
2019	1608.56	1471.19	27.68	27.13	1040.81	927.43
2020	1563.7	1570.47	23.14	23.28	833.71	887.22
2021	1721.06	1676.45	18.72	19.6	860.68	848.44
2022	1767.7	1789.58	18.14	16.09	758.46	811.04
2023	1837.19	1910.34	14.06	12.74	764.6	774.95
2024	–	2039.25	–	9.54	–	740.15
2025	–	2176.87	–	6.49	–	706.58
2026	–	2323.76	–	3.57	–	674.2

Table 2b. Prediction result of GM(1,1) for energy demand.

Years	Natural gas users (10^4 people)		Artificial gas users (10^4 people)		LPG users (10^4 people)	
	Actual value	Predicted value	Actual value	Predicted value	Actual value	Predicted value
2014	25972.94	25972.94	1757.01	1757.01	14378.41	14378.41
2015	28561	29900.11	1322	1217.18	13955	14263.38
2016	30856	31779.42	1085	1065.08	13744	13428.44
2017	33934	33776.85	752	923.79	12616	12642.39
2018	36902	35899.83	779	792.52	11782	11902.34
2019	39025	38156.24	675	670.58	11297	11205.62
2020	41302	40554.47	548	557.3	10767	10549.67
2021	44195.53	43103.43	455.77	452.06	10180.49	9932.13
2022	45678.51	45812.61	380.51	354.3	9332.83	9350.74
2023	47114.77	48692.07	312.3	263.48	8405.8	8803.37
2024	–	51752.5	–	179.11	–	8288.05
2025	–	55005.3	–	100.73	–	7802.9
2026	–	58462.54	–	27.91	–	7346.14

Table 3. Residual ratio.

Year	Natural gas supply (10^8 m^3)	Natural gas users (10^4 people)	Artificial gas supply (10^8 m^3)	Artificial gas users (10^4 people)	LPG supply (10^4 tons)	LPG users (10^4 people)
2014	0	0	0	0	0	0
2015	−0.08859	−0.04689	0.057488	0.079289	−0.06191	−0.0221
2016	−0.0322	−0.02993	0.097301	0.018359	0.020106	0.02296
2017	−0.02161	0.004631	−0.30602	−0.22844	−0.01353	−0.00209
2018	0.045542	0.027158	−0.04599	−0.01736	0.045522	−0.01021
2019	0.085399	0.022262	0.01987	0.006548	0.108934	0.008089
2020	−0.00433	0.018099	−0.00605	−0.01697	−0.06418	0.020185
2021	0.02592	0.024711	−0.04701	0.00814	0.014221	0.024396
2022	−0.01238	−0.00294	0.11301	0.068881	−0.06932	−0.00192
2023	−0.03982	−0.03348	0.093883	0.156324	−0.01354	−0.0473

Table 4. Divide intervals and corresponding states.

Interval /states D_p	Natural gas supply (10^8 m^3)	Natural gas users (10^4 people)	Artificial gas supply (10^8 m^3)	Artificial gas users (10^4 people)	LPG supply (10^4 tons)	LPG users (10^4 people)
D_1	[−0.0886, −0.0306]	[−0.0469, −0.0222]	[−0.3060, −0.1663]	[−0.2284, −0.1002]	[−0.0693, −0.0099]	[−0.0473, −0.0234]
D_2	(−0.0306, 0.0274]	(−0.0222, 0.0025]	(−0.1663, −0.0267]	(−0.1002, 0.0281]	(−0.0099, 0.0495]	(−0.0234, 0.0005]
D_3	(0.0274, 0.0854]	(0.0025, 0.0272]	(−0.0267, 0.1130]	(0.0281, 0.1563]	(0.0495, 0.1089]	(0.0005, 0.0244]

ratio was classified according to which interval they belong to, and the state transition matrix for each sequence was derived. Additionally, both the original residual intervals (states) and predicted residual intervals (states) are included.

Next, we corrected the GM prediction results based on the predicted residual states, and the curve of corrected results are shown below (Figure 4).

It is evident from the calculated prediction results that the curve obtained by the GM–Markov hybrid forecasting method is more consistent with the actual values. Compared with the use of only GM, prediction accuracy has been significantly improved, with line modified by Markov is closer to the actual data line.

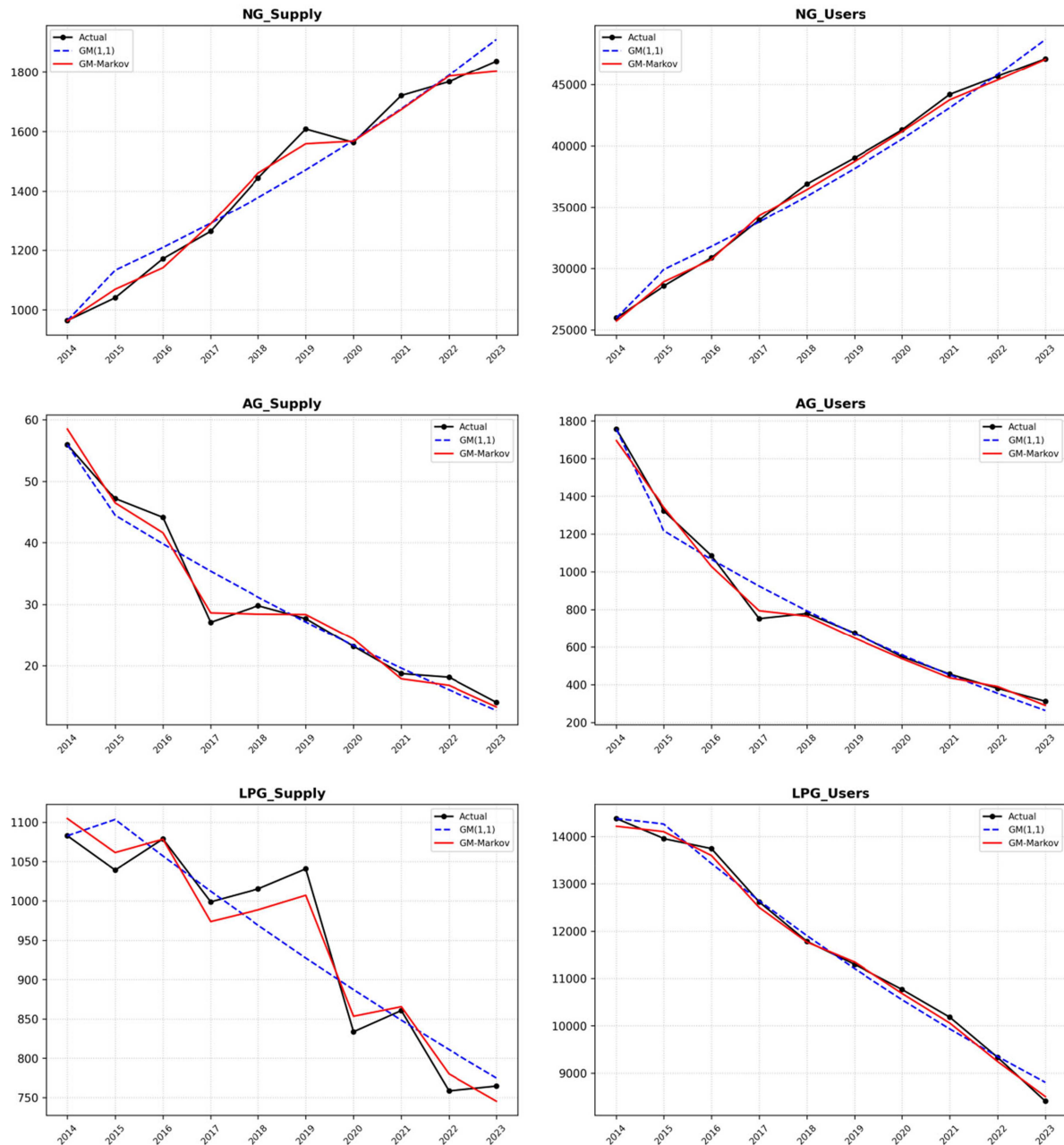


Figure 4. GM, GM–Markov results in NG, AG, LPG supply and users.

4.2. Parameter searching with WOA

As previously observed, GM–Markov integrated method shows significant improvement over traditional GM(1,1). To further explore optimal parameter settings and enhance model accuracy, we introduced Whale Optimization Algorithm (WOA), a swarm intelligence search algorithm, from a series of search algorithms. Table 5 below shows the parameter setting.

The population size of 30 was selected, because smaller populations of 10 or 20 occasionally converged to suboptimal solutions due to insufficient exploration, whereas larger populations offered limited improvement in fitness while increasing computational cost. Thirty individuals provided a balance between search adequacy and efficiency. Through the above steps, we developed a program and conducted experiments, with the optimized parameters presented in Table 6 below:

Based on the weight data, it is found that the prediction for AG Supply Group relies more on the lower bound of its range, while the remaining groups rely more on the upper bound (with the lower bound having a larger weight).

After obtaining the parameters, we incorporated the parameters into our forecasting model and recalculated the forecasting results. To visualize the observation of the development trends, we plotted the forecasting results of three methods (GM, GM–Markov and GM–Markov–WOA) with original data in Figure 5.

Based on the results, we also incorporated the forecasting results optimized by WOA into the comparison. It can be observed that, compared with the model without parameter optimization, the WOA-optimized model exhibits higher accuracy.

It should be noted that the first point of the predicted value after Markov correction will have a difference from the original value. This is because the original error is converted into an interval, and then the prediction error is calculated based on that. This is the reason for the difference. Existing research has confirmed the rationality of this characteristic (Xu, Lin, and Du 2024).

4.3. Analysis of the results

In this section, we will explain the trends of the data and conduct analysis in the field of urban energy policy.

Natural Gas (NG) dominates current urban gas consumption, with both its supply and user population exhibiting steady growth. Forecast results indicate that by 2026 China's urban natural gas supply will reach approximately 225.5 billion cubic meters, while the number of users will expand to roughly 567.4 million. Such sustained expansion implies that urban pipeline networks and gas storage facilities must maintain a moderately proactive construction pace. For northern heating regions and major population destinations such as the Chengdu–Chongqing area and the Yangtze River Delta, winter supply pressure will remain difficult to be released unless peak-shaving capacity increases in step with user growth. The high fitting accuracy of the WOA–GM–Markov model on the demand side further suggests that, despite adjustments to the coal-to-gas policy, the rigid demand from residential and commercial users remains strong. Policymakers should therefore ensure orderly expansion of trunk pipelines and LNG terminals in the near term, even while advancing electrification and pilot renewable gas projects, to avoid localized supply–demand imbalances arising from underestimation. This finding isn't conflict to the dual-carbon

Table 5. Parameter settings for (WOA).

Parameter	Symbol	Value	Description
Population size	N	30	Number of search agents (whales)
Maximum iterations	$T_{\max}$	100	Termination condition for optimization
Search range	φ_i	[0,1]	Bounds for the residual weight parameter
Convergence parameter	a	$2 \rightarrow 0$	Linearly decreases with iteration t (Eq. (21))
Number of independent runs	–	10	Runs repeated to mitigate random initialization effect, and retain best results

Table 6. Optimization parameter results φ_i

Group	NG supply	NG users	AG supply	AG user	LPG supply	LPG users
φ_i	0.1655	0.3334	0.7416	0.2181	0.3617	0.0152

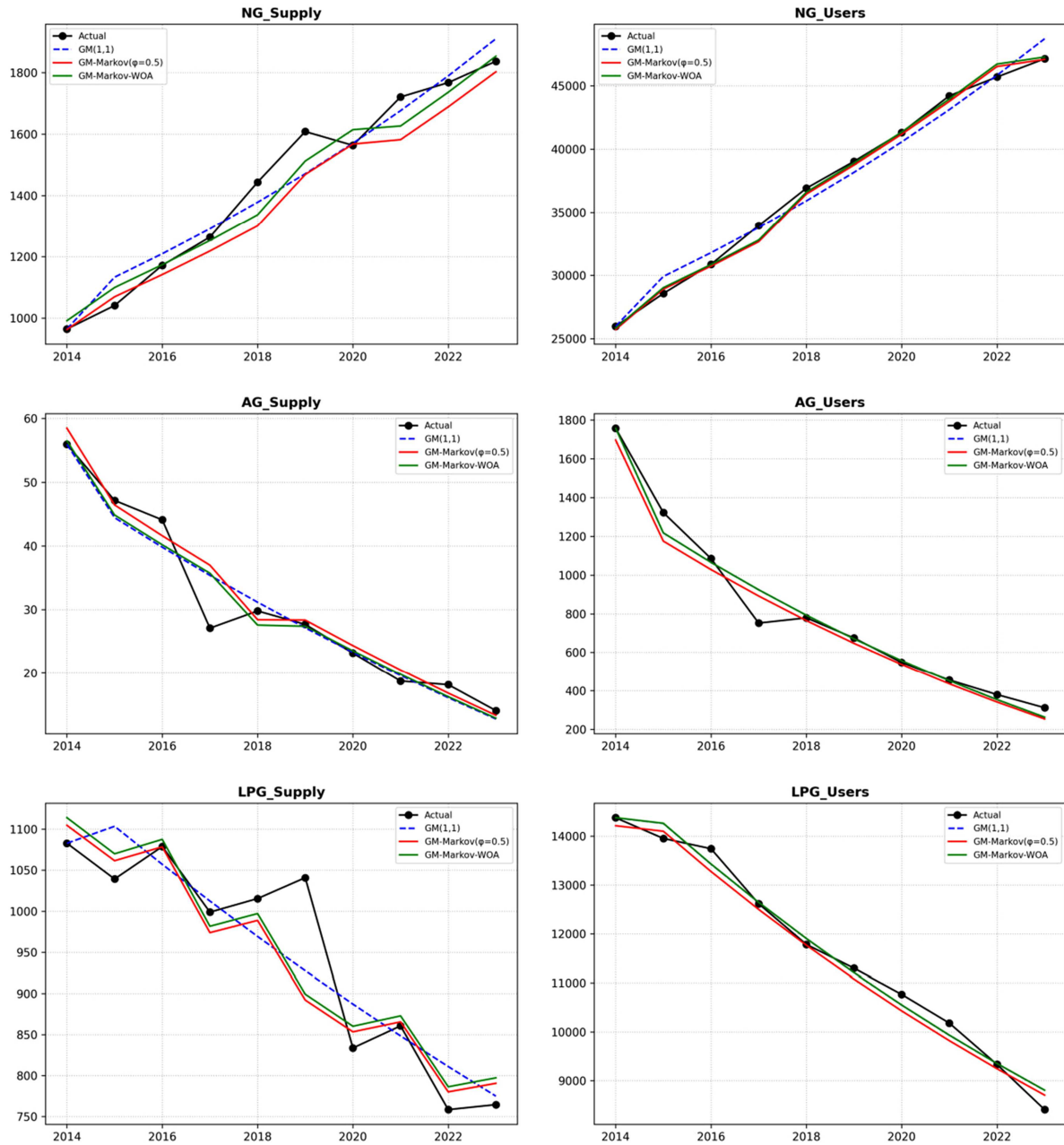


Figure 5. GM, GM-Markov, GM-Markov-WOA results comparison in NG, AG, LPG supply and users.

goals, cause stable natural gas supply serves as a critical transitional measure to ensure urban energy system security before large-scale renewable deployment matures.

The forecast for Artificial Gas (AG) points to a structural decline. Supply volume is projected to fall steadily from 1.2861 billion cubic meters in 2023 to about 0.3604 billion cubic meters by 2026, accompanied by a concurrent shrinkage in the user base. The WOA-GM-Markov accurately tracks this downward trajectory. From a policy perspective, the phase-out of AG represents not only an outcome of constraints on high-polluting fuels, but also a timeline requirement for repurposing existing urban pipeline assets (Huang et al. 2023). Urban planning authorities should rely on precise estimates of user population to decommission old gas pipelines or convert them into natural gas distribution branches. For the limited number of aging residential communities and specific industrial users still dependent on artificial gas, governments should prepare alternative energy access plans and financial subsidy schemes ahead of time, to minimize sunk costs while ensuring basic energy needs.

Liquefied Petroleum Gas (LPG) functions as a transitional fuel in this study. Projections show that its supply volume and user population will remain relatively stable over the next three years, declining at a slower pace than artificial gas. In counties, townships, and urban migrant-concentrated areas not yet served by natural gas pipelines, LPG will remain an important supplementary source due to its flexible storage and transportation and low infrastructure costs. Urban energy authorities therefore need not rush to eliminate LPG entirely. Instead, they should dynamically optimize the distribution of bottled gas supply sites based on the forecast data from this study, ensuring sufficient and safe LPG availability in pipeline coverage gaps. Meanwhile, as pipeline networks gradually extend through built-up urban districts, LPG demand will naturally attenuate. Such a smooth transition helps avert the emergence of energy poverty.

Proposed model not only enhances prediction accuracy under small-sample conditions but also provides decision-making support for differentiated urban energy policy design. For high-growth categories, emphasis should be placed on proactive infrastructure planning. For declining categories, on orderly phase-out and asset disposal and for transitional categories, on supply equity and resilience. This data-driven governance approach can assist cities in achieving a dynamic balance between energy security and low-carbon transition under dual-carbon constraints.

5. Model validation

Based on the above experiments, proposed GM–Markov–WOA is generally effective for short time-series forecasting. However, its robustness and predictive performance require further validation, especially compared with another common model: ARIMA and other sophisticated optimization algorithms. Moreover, parameter variations will influence the outcomes, so the forecasting results under different parameters also need to be verified.

5.1. Robustness test

Given the limited sample size ($n = 10$), it is essential to verify that the proposed model maintains predictive stability under out-of-sample conditions rather than merely fitting historical sequence. To this end, a rolling-origin validation was conducted. A fixed training window of six years was adopted, and the model was used to forecast the immediately subsequent year. The window advanced annually, yielding four independent one-step-ahead predictions for 2020–2023. In each iteration, the entire WOA–GM–Markov pipeline was executed independently. Table 7 compares the rolling-window RMSE with the full-sample RMSE for all six sequences.

Several observations emerge from Table 7. Two artificial gas sequences (AG Supply and AG Users) achieve rolling RMSE values lower than their full-sample counterparts, with ratios of 59.9% and 59.1%, respectively. These series follow a persistent exponential decline, and the result demonstrates that the framework captures structural trends for that reducing the training data from ten to six does not degrade predictive accuracy. This constitutes strong evidence against overfitting, as a model merely memorizing historical noise would exhibit substantially inflated out-of-sample errors on a smaller training set.

For the largest and most policy-relevant sequence, NG Users, the rolling RMSE ratio is 158.0%, remaining well within a factor of two of the full-sample fit. Given that the training window contains only 60% of the available data, this moderate increase confirms that the model retains meaningful predictive capacity.

The higher ratios observed for NG Supply (191.5%), LPG Supply (194.1%), and LPG Users (185.7%) can be interpreted in light of the small number of validation points. Only four out-of-sample predictions

Table 7. Comparison between rolling-origin validation and full-sample fitting.

Energy sequence	Full-sample RMSE	Rolling RMSE	RMSE ratio
NG supply	61.04	116.90	191.5%
NG users	782.64	1236.29	158.0%
AG supply	3.25	1.95	59.9%
AG users	75.57	44.69	59.1%
LPG supply	50.53	98.10	194.1%
LPG users	220.68	409.88	185.7%

obtainable from a ten-year series and a six-year window. For LPG Supply, the 2020 prediction error of 170.41 contrasts sharply with 34.36, 39.22, and 82.09 in the subsequent three years (see Figure 5), confirming that the elevated mean is driven by a single volatile year rather than systematic model failure. Under normal post-2020 conditions, the model's predictive accuracy is considerably higher.

In summary, the rolling-origin validation demonstrates that the model maintains acceptable out-of-sample stability. The model performs best on trend-consistent sequences and shows degradation primarily for pandemic year, which is a reasonable limitation for any data-driven forecasting approach.

5.2. Performance comparison

To further validate the superiority of the proposed WOA–GM–Markov model, two additional benchmark methods were introduced for comparison, ARIMA and a PSO-optimized GM–Markov. ARIMA (Autoregressive Integrated Moving Average) is a widely used statistical method for time series forecasting (A, Christo, and Elizabeth 2025). While Particle Swarm Optimization (PSO) was employed as an alternative method to optimize parameter ϕ_l in GM–Markov framework. PSO was configured with a swarm size of 30 and 100 iterations, identical to the WOA setup to ensure fair comparison.

Five methods, GM(1,1), ARIMA, GM–Markov ($\lambda = 0.5$), PSO-optimized GM–Markov, and WOA-optimized GM–Markov, were evaluated, and the results for RMSE and MAE are presented in Figures 6 and 7 respectively. To illustrate the differences in error magnitude among the methods, a logarithmic scale was adopted for the vertical axes of both charts.

The prediction errors of the GM–Markov hybrid models were substantially lower than those of the two conventional benchmarks. Taking NG Users as an example, RMSE dropped from 932.04 (GM(1,1)) to 551.93 (GM–Markov, $\phi = 0.5$), and further to 529.46 under both optimized versions, a total reduction of 43.2%.

A direct comparison reveals WOA's consistent advantage. On NG Supply, WOA achieved an RMSE of 61.35, marginally outperforming PSO (61.36). On the remaining five sequences, the two optimizers converged to equivalent solutions. Notably, for AG Supply, the standard GM–Markov ($\phi = 0.5$) produced an RMSE of 3.46, worse than GM(1,1) alone (3.22), demonstrating that midpoint selection may also degrade predictive accuracy. The WOA-optimized version restored and slightly improved performance to 3.26, confirming its value of principled parameter optimization over subjective assignment.

ARIMA performed poorly under this small-sample conditions. For NG Users, its RMSE of 18,072.54 was approximately 34 times that of WOA–GM–Markov. For AG Supply, ARIMA reached 18.78, over five

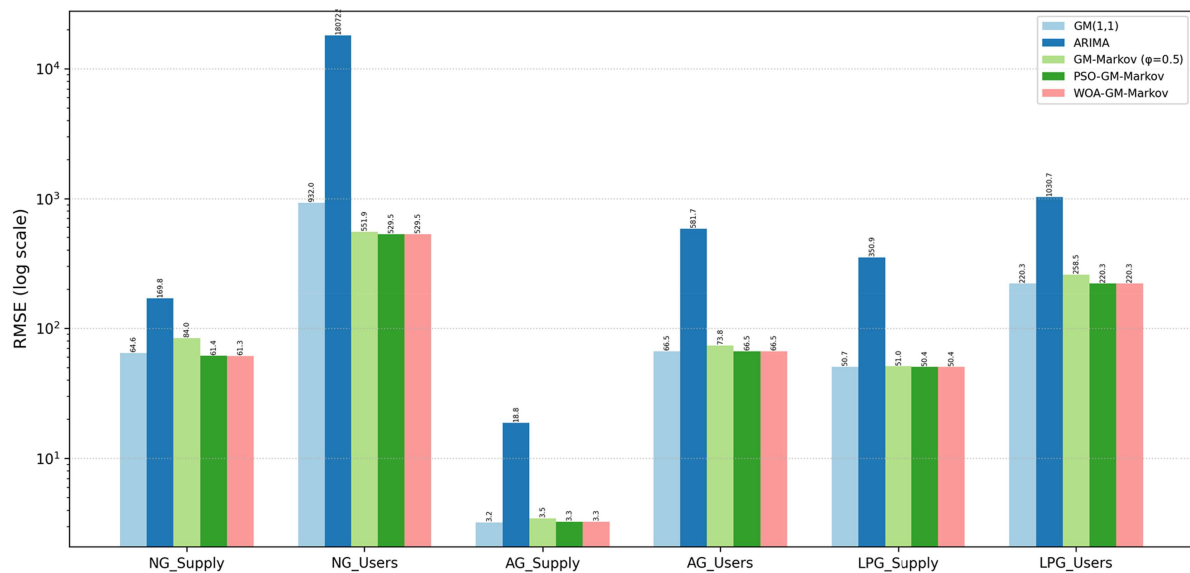


Figure 6. Comparison of RMSE across 5 models.

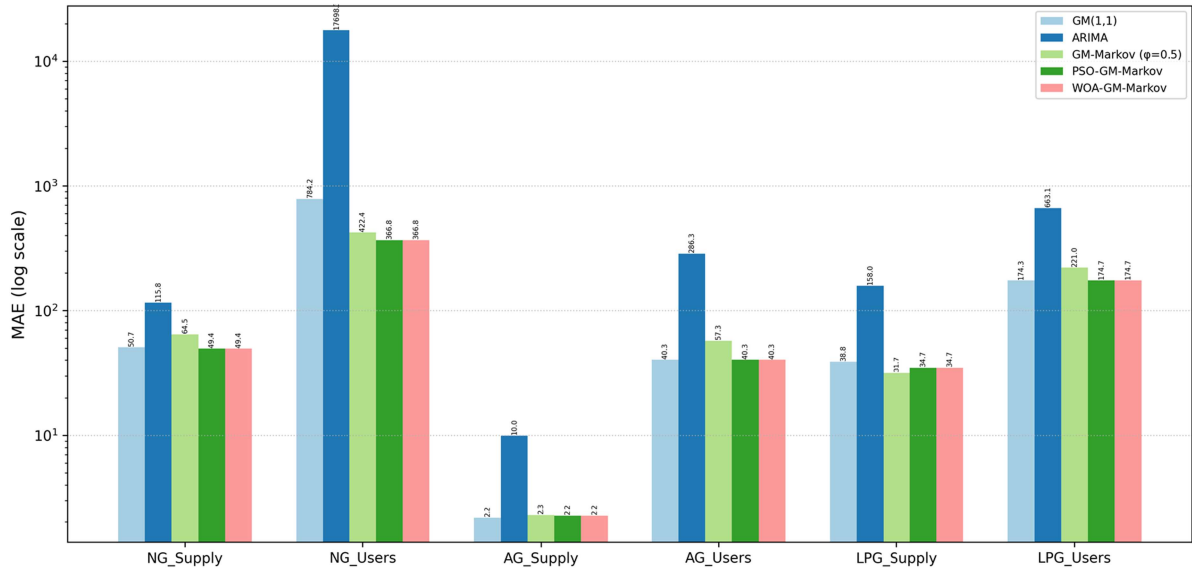


Figure 7. Comparison of MAE across 5 models.

times the optimized models. This limitation stems from ARIMA's reliance on stable autocorrelation structures, which become unreliable with only ten observations.

In summary, WOA–GM–Markov achieves meaningful accuracy gains through systematic parameter optimization and is a more suitable choice than ARIMA for small-sample urban gas forecasting. The optimization procedure eliminates the performance degradation risk inherent in GM–Markov ($\phi = 0.5$), and WOA's mechanism justifies its selection as the embedded optimizer.

6. Conclusion

This study aims to improve the accuracy of short-term supply–demand forecasting for urban primary energy (including NG, AG, LPG), addressing the limitations of traditional models' poor adaptability to volatility and GM–Markov's subjective parameter setting. We integrate Grey System Theory, Markov Chain, and WOA to construct WOA–GM–Markov for prediction, which is verified using 2014–2023 data from China's National Bureau of Statistics. We have certain findings below.

- NG (high-growth, low-carbon fuel): WOA–GM–Markov significantly mitigates demand volatility caused by urbanization and coal-to-gas policies.
- AG (fuel energy): the model accurately fits the stable downward trend of supply and user attrition, and the downward trend is due to environmental regulations.
- LPG (stable transitional fuel): the model maintains high accuracy, supporting policymaker balancing supply chains for underserved urban areas.

The core innovation and contribution of this research is reflected in the following three aspects:

- Theoretical value:** GM–Markov, which integrates the trend forecasting capability of grey models and the fluctuation forecasting advantage of Markov models, is applied to short-term discrete time series forecasting. By incorporating WOA, it addresses the over-subjectivity in parameter setting of existing models, enriching the theory in this field.
- Practical value:** research on short-term forecasting of primary energy in urban areas remains scarce, for that most studies focusing on machine learning (Lahyani and Amayri 2025; Mohammed et al. 2025; Renold, Sinha, and Gao 2025; Riquelme-Dominguez et al. 2024; Uceta-Acosta et al. 2025) and few on traditional forecasting approaches. According to the dataset obtained from National Bureau of Statistics

of China from 2014 to 2023, the resulting supply–demand forecasts can serve as a quantitative basis for implementing policies related to urban energy supply security, demand-side management, and emergency responses, helping policymakers identify future trends and finding out opportunities in advance.

- C. **Application value:** by introducing energy-consuming population as an indicator of energy demand, this study deeply links population-related information with energy supply and demand. The forecasting results are not only applicable to energy sectors but also provide references for urban planning departments and industrial supervision authorities, facilitating the synergistic linkage of multi-departmental policies.

In the related fields of the future, we offer the following directions that can be further developed continuously.

- A. Optimize GM's parameters. current GM uses the average for adjacent mean $z^{(1)}$ and least squares for unbiased estimation of a (development parameter) and b (grey action quantity). Future work can explore alternative formulations for $z^{(1)}$ (weighted average based on data volatility) and apply intelligent several algorithms to optimize a and b and make comparison, further enhancing GM's fitting ability for non-linear data.
- B. Compare multiple optimization algorithms. extend other searching and optimization model include PSO or GWO. For each forecasting model, calculate MSE of models optimized by different algorithms to identify the most suitable optimizer, improving the model's adaptability to diverse energy dynamics.

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Qiyong Yang: methodology, validation, writing—original draft, visualization; Jidan Huang: conceptualization, methodology, writing—review and editing; Tao Dai: funding acquisition, writing—review and editing.

Author contributions

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Appendix

Table A1. Prediction result by GM–Markov–WOA.

Year	Natural gas supply (10^8 m ³)	Natural gas users (10^4 people)	Artificial gas supply (10^8 m ³)	Artificial gas users (10^4 people)	LPG supply (10^4 tons)	LPG users (10^4 people)
2014	991.548	25824.805	56.481	1757.220	1114.061	14380.362
2015	1099.350	29016.946	44.852	1217.325	1069.979	14265.316
2016	1173.540	30840.746	40.178	1065.207	1087.579	13430.263
2017	1252.727	32779.178	35.716	923.900	981.516	12644.106
2018	1337.270	36593.789	27.570	792.615	997.043	11903.956
2019	1512.636	38893.816	27.388	670.660	899.209	11207.141
2020	1614.713	41338.405	23.501	557.367	860.222	10551.102
2021	1626.674	43936.637	19.786	452.114	872.895	9933.478
2022	1736.445	46698.187	16.243	354.342	786.360	9352.009
2023	1853.619	47253.844	12.861	263.511	797.286	8804.565
2024	1978.702	50223.878	9.631	179.131	761.483	8095.662
2025	2112.236	53380.599	6.552	100.742	685.079	7621.774
2026	2254.764	56735.722	3.604	27.913	693.632	7175.617

Table A2. Best move_{opt}.

Energy sequence	Natural gas supply	Natural gas users	Artificial gas supply	Artificial gas users	LPG supply	LPG users
move _{opt}	0	0	58.23	924.61	207.61	0

Table A3a. RMSE of 6 sequences and 5 prediction methods.

Sequence	GM(1,1)	ARIMA	GM–Markov	PSO–GM–Markov	WOA–GM–Markov
NG_Supply	64.6	169.77	83.98	61.35	61.35
NG_Users	932.04	18072.54	551.93	529.46	529.46
AG_Supply	3.22	18.78	3.46	3.26	3.26
AG_Users	66.53	581.66	73.78	66.53	66.53
LPG_Supply	50.66	350.9	50.98	50.38	50.38
LPG_Users	220.34	1030.74	258.5	220.34	220.34

Table A3b. MAE of 6 sequences and 5 prediction methods.

Sequence	GM(1,1)	ARIMA	GM–Markov	PSO–GM–Markov	WOA–GM–Markov
NG_Supply	50.68	115.75	64.49	49.43	49.43
NG_Users	784.16	17698.29	422.37	366.76	366.76
AG_Supply	2.16	9.95	2.27	2.24	2.24
AG_Users	40.25	286.34	57.28	40.25	40.25
LPG_Supply	38.78	158.04	31.72	34.72	34.72
LPG_Users	174.32	663.05	221	174.69	174.68

Table A4. Detailed rolling-origin validation results for all sequences.

Sequence	Training window	Predicted year	Actual value	Predicted value	Absolute error
NG supply	2014–2019	2020	1563.70	1790.32	226.62
	2015–2020	2021	1721.06	1749.87	28.81
	2016–2021	2022	1767.70	1817.36	49.66
	2017–2022	2023	1837.19	1840.87	3.68
NG users	2014–2019	2020	41302.00	42580.20	1278.20
	2015–2020	2021	44195.53	45063.57	868.04
	2016–2021	2022	45678.51	46916.45	1237.94
	2017–2022	2023	47114.77	48595.93	1481.16
AG supply	2014–2019	2020	23.14	20.40	2.74
	2015–2020	2021	18.72	18.69	0.03
	2016–2021	2022	18.14	20.63	2.49
	2017–2022	2023	14.06	15.30	1.24
AG users	2014–2019	2020	548.00	503.33	44.67
	2015–2020	2021	455.77	462.26	6.49
	2016–2021	2022	380.51	457.36	76.85
	2017–2022	2023	312.30	318.91	6.61
LPG supply	2014–2019	2020	833.71	1004.12	170.41
	2015–2020	2021	860.68	895.04	34.36
	2016–2021	2022	758.46	797.68	39.22
	2017–2022	2023	764.60	682.51	82.09
LPG users	2014–2019	2020	10767.00	10628.29	138.71
	2015–2020	2021	10180.49	9890.80	289.69
	2016–2021	2022	9332.83	9721.23	388.40
	2017–2022	2023	8405.80	9052.33	646.53